transparent manner, which facilitates the transfer of critical technologies. It has also been pointed out that though the MQ-9B is used primarily for ISR (intelligence surveillance & reconnaissance) activity, it can easily be armed with a formidable range of weaponry. The MQ-9Bs will provide India with an unmatched capability to keep watch over the Line of Actual Control (LAC) and the Line of Control (LoC) which it had hitherto lacked. The joint statement highlighted that General Atomics will set up a maintenance, repair and overhaul (MRO) facility in India to support New Delhi's goal of enhancing indigenous defence capabilities. The firm is also reportedly in contact with several Indian companies for the domestic manufacture of components.

THE FORMIDABLE MQ-9B

The MQ-9B—an updated version of the MQ-9A Reaper—is designed for a variety of roles, including maritime surveillance, anti-submarine warfare (by dropping sonobuoys—a floating cylindrical sonar system), over-the-horizon targeting (the ability to detect targets beyond the range of ordinary radars) and long-range strategic ISR using its Lynx multi-mode radar. With a maximum range of 11,112 km and maximum endurance (the ability to be airborne without intervention) of 40 hours, MQ-9B SkyGuardians and SeaGuardians can be used for anti-surface warfare (by launching missiles), airborne mine countermeasures (by using sonar to detect mines), airborne early warning as well as for search and rescue operations.

The MQ-9B can also conduct precision-guided ground bombing with GBU-12 Paveway II laser-guided bombs and GBU-38 Joint Direct Attack Munition (JDAM) kits. Armed with AIM-9 sidewinder missiles and AIM-92 Stinger air-to-air missiles, it can engage helicopters, UAVs and other airborne targets. The strike capabilities of the Reaper and its predecessor, the RQ-1/ MQ-1 Predator, are well known. In September 2022, the US military used an MQ-9A Reaper to execute Al Qaeda leader Ayman al-Zawahiri in Kabul. In January 2020, a Predator strike killed Iranian general Qassem Soleimani in Baghdad.

Indian military planners claim that with armed drone capability, India will be able to launch remote-controlled operations and surgical strikes, such as on terrorist hideouts in Pakistan-occupied Kashmir, and engage targets on the Himalayan borders with China. Since the aggression of the Chinese military on the LAC in 2020, the navy has deployed two MQ-9As leased from GA for ISR, while the army has been using the indigenous SWITCH tactical UAV made by ideaForge to boost its surveillance capabilities.

Lt. Gen. Vinod Bhatia, a former director-general of military operations, says the MQ-9B drones will exponentially enhance India's ISR capabilities, both along the LAC and in the maritime domain. "The long-range high endurance seek-and-shoot capability gives an unmatched operational advantage to our armed forces," he says. Another officer claims that India's procurement of MQ-9Bs will help offset the advantage Pakistan and China currently have with Turkish-made Bayraktar TB2, Wing Loong 3 and Cai Hong 4 armed drones.

Air Marshal S.B.S. Sinha, former deputy chief of the IAF, says that the Kargil intrusion in 1999 and the Galwan incident in 2020 happened due to a lack of electronic surveillance capabilities. "MQ-9Bs will help prevent such surprises," he says. Another IAF officer adds that MQ-9Bs will secure Indian airspace and complement the IAF's Phalcon Airborne Early Warning and Control Systems (AWACS), and NETRA Airborne Early Warning and Control System (AEW&CS) aircraft. The MQ-9Bs integrate with US-origin platforms operated by India, like P-81 long-range maritime patrol aircraft, Apache attack helicopters and MH-60 helicopters.

MQ-9Bs will complement the IAF's AWACS and NETRA aircraft, and reduce the workload of the Navy's P-81 patrol aircraft

With MQ-9Bs able to undertake much of the workload of a maritime patrol aircraft at a fraction of its cost, the navy is keen to reduce the expense and wear and tear involved in operating its fleet of 12 P-8Is. It's a fact readily attested to by Rear Admiral Sudarshan Shrikhande (retd). "It will enable P-8I aircraft to devote more time to train for anti-submarine warfare and keep track of 'unfriendly' submarines in the Indian Ocean and beyond," he says.

THE REAL DEAL

After India joined the Missile Technology Control Regime (MTCR) in 2016 and became eligible to procure long-range drones like the MQ-9B, India and the US have been negotiating on a drone deal. The Trump administration had cleared the sale of SeaGuardians to India. Yet the talks went cold—possibly due to the high costs, doubts by the armed forces and concerns about the Make in India content. However, till November 2022, the two MQ-9As

MQ-9B
GUARDIANS OVER OUR SKY

With its long-endurance surveillance capability, SkyGuardians and SeaGuardians (in pic) will be the military's eyes. An MQ-9B's chief features include:

- 240 Knots True Air Speed (KTAS)
- 40 hours Maximum Endurance
- 40,000 ft MSL (mean sea level) service ceiling
- 11,112 km maximum range
- 1,746 kg payload capacity
- Operates 'on the horizon'